

APEX INSTITUTE OF TECHNOLOGY & SCIENCES (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

MIDTERM EXAMINATION EXAMINATION • AY 2025-26

Course: Operating Systems (CS502) | Faculty: Dr. Rajesh Menon

Date: 18 October 2025

Time: 90 Minutes

Max Marks: 30

Instructions: Answer all questions in Part A. Part B contains internal choice.

PART A — CONCURRENCY & DEADLOCKS

Answer all questions. (30 Marks)

1. Consider 5 processes P0 to P4 with Allocation and Max matrices. Check whether the system is in a safe state using Bankers Algorithm. [15 M]
2. Implement Readers-Writers problem using counting semaphores ensuring writer starvation is prevented. [15 M]